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$$\begin{aligned} & \lim_{x \rightarrow -\frac{3}{2}} \left(\frac{2x+3}{4x^2+12x+9} \right) \\ &= \frac{2\left(-\frac{3}{2}\right)+3}{4\left(-\frac{3}{2}\right)^2+12\left(-\frac{3}{2}\right)+9} \\ &= \frac{-3+3}{9-6\cdot 3+9} \\ &= \frac{0}{0} \\ & \lim_{x \rightarrow -\frac{3}{2}} \left(\frac{2x+3}{(2x+3)^2} \right) = \lim_{x \rightarrow -\frac{3}{2}} \left(\frac{1}{2x+3} \right) \\ & \lim_{x \rightarrow -\frac{3}{2}-} \left(\frac{1}{2x+3} \right) = \frac{1}{0^-} = -\infty \\ & \lim_{x \rightarrow -\frac{3}{2}+} \left(\frac{1}{2x+3} \right) = \frac{1}{0^+} = +\infty \end{aligned}$$

References